

Introduction – Neuroaesthetics

Neuroaesthetics is a young interdisciplinary research area aimed at exploring the neuro-physiological and evolutionary basis of our aesthetic experience both in the production and fruition of beauty and artworks. It offers a privileged and peculiar field of encounter between Science and Art, especially in what concerns the discussion on the status of visual images, from the perception related to object recognition to its mental symbolic unfolding, or its role in the production of knowledge. In fact, most works in neuroaesthetics (NE) focus on visual art and one of the most powerful instruments for NE studies is neuroimaging. That is why the research project “Image in Science and Art” of the Center for Philosophy of Sciences of the University of Lisbon (CFCUL) could not omit promoting a discussion on this controversial field of research.

Generally accused of reductionism, its epistemological premises and methods are not clearly defined yet. Indeed, we might recognize two main approaches to neuroaesthetics: two main approaches to NE: a more reductionist one (strongly supported on neuroimaging), which basically considers the artistic experience in terms of stimuli of specific areas of the brain, and another that tries to explain art on the basis of a broader conception of the brain as embedded in a body and in an environment.

The first one aims at identifying the neural correlates of artistic experience, that is, which brain areas have to be activated in order to engender the aesthetic experience in human beings. In this kind of studies it is sometimes implied that if there are specific areas of the brain involved in the aesthetic appreciation, then human beings produce art in order to please the brain (that of the artist and that of the beholder). Furthermore it is assumed that we appreciate art because it excites some specific areas of our brain¹. This approach relies essentially on a more reductionistic epistemological framework, using anatomo-metabolical data provided by functional Magnetic Resonance Imaging (fMRI), a technique that visualizes which areas of the brain are activated when a subject is engaged in specific tasks.

In the second group we can gather different stances, which are not mutually exclusive. One is the evolutionary approach, in which artistic creation and aesthetic pleasure are investigated as possible products of the evolution of the brain in the

¹ It is worth to notice that in neuroaesthetics literature the terms “artistic experience” and “aesthetic experience” are generally used as synonyms, although this equivalence is not justified.

struggle for survival. Another line of research could be labelled as “pathological approach”, since it studies art production as the possible result of some neuro-pathological condition, with examples ranging from Van Gogh’s alleged schizophrenia, to de Kooning’s Alzheimer, or the so-called *savant syndrome* (the proposed underlying condition that could explain some of the drawings made by autistic children). Finally, a different line of investigation looks at art as an embodied experience. This approach, which has its philosophical antecedents in 19th century German theories on empathy and in Merleau-Ponty’s phenomenology, has recently received a strong impulse thanks to the discovery of mirror neurons.

As shown by the very short (and necessarily incomplete) review here published, neuroaesthetics is far from being a well-defined discipline. On the contrary, we face a recent and very much controversial set of research programs whose explicative ambition cannot leave us indifferent, since it defeats our conception of art as one of the most sublime expressions of human creativity. Aiming at contributing to the debate about these researches, we organized, in March 2009, a two days workshop intended to discuss neuroaesthetics and its ability to explain the aesthetic experience. The success of the conference made us realize how great and extended is the curiosity about neuroaesthetics, especially among researchers and students from biology, cognitive sciences, aesthetics and philosophy of science. This gave us the impulse for editing a book that might be considered a “critical introduction” to neuroaesthetics. With this objective in mind we decided to publish, together with some of the papers presented at the workshop, the reprint of three seminal articles in neuroaesthetics.

The volume is thus organized in two parts. In the first part the reader will find the reprint of three articles that epitomize different neurological approaches to aesthetics, namely, *The Science of Art. A Neurological Theory of Aesthetic Experience*², by Ramachandran and Hirstein, *Neural correlates of beauty*³, by Kawabata and Zeki, and *Motion, emotion and empathy in aesthetic experience*⁴, by Freedberg and Gallese. In the second part we have collected some of the papers presented in our workshop, namely those by Chiara Cappelletto and Andrea Pinotti, from the Department of

² V.S. Ramachandran and W. Hirstein, *Journal of Consciousness Studies*, 1999; 6(6-7):15-51.

³ H. Kawabata and S. Zeki, *Journal of Neurophysiology*, 2004; 91:1699-1705.

⁴ D. Freedberg and V. Gallese, *Trends in Cognitive Sciences*, 2007; 11(5):197-203.

Philosophy of the State University of Milan, and by Vasco Correia, Alexander Gerner and Catarina Pombo Nabais, from the Center for Philosophy of Sciences of the University of Lisbon.

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The opening article is *The Science of Art. A Neurological Theory of Aesthetic Experience* published in 1999 by **V.S. Ramachandran and William Hirstein** in the *Journal of Consciousness Studies*. In this work, generally considered a founding text of the “evolutionary branch” of NE, the stated aim of the authors is to give an empirical answer to the question of how our brain responds to art, in a coherent biological framework framework that encompasses the internal logic of art (its “universal laws” or “principles” – albeit mainly in a perceptual only approach to art), an evolutionary rationale of these principles and the neurological bases of these phenomena.

Ramachandran and Hirstein propose eight universal rules, «analogous to the Buddha’s eightfold path to wisdom»: the *peak shift effect*, a well studied psychological phenomenon involved in the recognition of and response to objects’ peculiar features; *grouping* and *isolating*, two complementary principles that allow, respectively, to extract specific objects from the visual field and to overlook redundant information; *contrast*, a way of reinforcing relevant information; *perceptual problem-solving*, *abhorrence of unique viewpoint*, *symmetry* and *visual metaphors*, a set of principles the authors define as a «mental tunnel between two concepts or percepts that appear grossly dissimilar in the surface».

According to the authors, all of these “laws” are related to evolutionary constraints – with particular reference to the allocation of attentional resources – and their neural bases are located in specific areas of the brain. What the artists do is to apply these rules *consciously* or *unconsciously*, in order to «optimally titillate» the visual brain. This reductionist approach is partially mitigated by Ramachandran and Hirstein’s admission that the art theory they suggest does not account for important matters like style or originality. Moreover, despite suggesting that «a great deal of what we call art is based on these eight principles» they recognize that «much of art is idiosyncratic, ineffable and defies analysis». In the second section of the book this article is widely commented by Cappelletto and Pinotti.

The second text, *Neural correlates of beauty*, was originally published by **Hideaki Kawabata and Semir Zeki** on the Journal of Neurophysiology, in 2004. In this article Zeki, who in his early works on NE claimed that «the function of art and the function of the visual brain are one and the same, or at least [...] the aims of art constitute an extension of the functions of the brain» (Zeki, 1999, p. 1⁵), seems to take a more cautious posture. In fact, rather than addressing the big “What is art?” question by means of functional Magnetic Resonance Imaging (fMRI) of the brain, he focuses on some terms of the problem that can be empirically investigated. The question addressed is whether there are (or not) areas of the brain that are specifically engaged when someone is watching a painting that he or she considers to be beautiful.

In this study ten adults with the same cultural background were asked to rate as beautiful, ugly or neutral a certain number of paintings. Afterwards they underwent fMRI during which they were shown the same paintings again. Results from this experiment show that there’s a correlation between certain aesthetic judgements and the activation of specific brain areas, and that, for example, the judgment of beautiful and ugly entails the activation of the same brain areas, although with different relative intensity. More specifically, the sight of a painting classified as beautiful or ugly engages principally the orbito-frontal cortex, involved in perception of rewarding stimuli and, interestingly enough, the motor cortex, which controls our motor functions. When a subject judges a painting beautiful the orbito-frontal cortex is more active than in the case of a painting judged ugly; conversely, the motor cortex is more active before ugly images than before beautiful ones. These areas, the authors remark, do not act in isolation. Not only they reciprocally interact, but they are also widely connected to other cortical regions, which «makes it likely that each can influence and be influenced by widespread regions of the cortex». One of these areas, the anterior cingulate, has often been associated with emotional states, which suggests a relationship between the aesthetic experience and emotions. In his article *Emotional Aspects of Neuroesthetics: A Naturalistic Approach*, Vasco Correia draws on Kawabata and Zeki’s article as a starting point to defend a hedonistic account of art.

After an article where emotion is somehow already taken into account (with some data implying a connection between the aesthetic sense and emotions), the third

⁵ S. Zeki, *Inner Vision. An exploration of art and the brain*, Oxford University Press, Oxford, 1999

article of the book is completely focused on the importance of emotion and corporeal sensation in the aesthetic experience.

The article, *Motion, emotion and empathy in aesthetic experience*, is a reprint from Trends in Cognitive Sciences. In this work, published in 2007, **David Freedberg and Vittorio Gallese** capitalize on the discovery of mirror neurons, which dates back to 2004, in order to challenge the idea that cognition plays a key role in the aesthetic experience. On the contrary, they propose that a crucial element of the response to art is to be found in empathy. In their view, empathy – generated by universal embodied mechanisms that encompass the simulation of actions, emotions and corporeal sensations – constitutes a basic level of response to all kind of images and, by extension, to artworks. They suggest that the empathetic response to images could rely on three features of our neuro-motor system located in specific regions of the brain: mirror neurons, canonical neurons and the sensorymotor response to others' emotions. The mirror neuron system, located in the ventral premotor cortex, is activated when we observe someone performing an action as if we were executing that very action and is also involved in the recognition of facial expressions. Canonical neurons, also located in the ventral premotor cortex, are activated when we observe static objects, as if we were actually manipulating and using them. Finally, the integrity of the sensorymotor system is known to be crucial for the recognition of emotions displayed by others.

Combining these elements it becomes possible to suggest a neurological account of a range of empathetic responses to representational visual art: the feeling of involvement in movements and emotions of others, the recognition of depicted emotions and the feeling of empathy for the represented bodily sensations. In what concerns abstract art, Freedberg and Gallese suggest that mirror neurons and the theory of embodied simulation could explain our reaction to the formal qualities of an artwork as a form of empathy towards the gestures that were performed by the artist in order to produce those specific marks. In this view, the empathetic relationship with an artwork is almost a precondition for further cognitive elaboration and does not collide with historical, cultural and other contingent factors that underpin our aesthetic experience. This article plays an important role in Cappelletto's and Pinotti's analysis.

Other seminal contributions on the relationship between aesthetics and the brain could have been presented in this volume. However Nevertheless, and as said

before, this book is not intended to exhaust all the existing perspectives on the subject, but to present the main views that somehow broke the way for others to follow.

As an example, we think it is worthwhile to mention is Terrence Deacon's view. Deacon is not exactly a neuroscientist, but an anthropologist with a strong biological and linguistics background. His research combines human evolutionary biology and neuroscience, with the aim of investigating the evolution of human cognition. However, unlike other scientists, he integrates the dimensions of language and symbol in the biological framework.

Deacon, who has written extensively on the evolution of human cognition from the perspective of semiotic processes⁶, presents in *The Aesthetic Faculty*⁷ (2006) some interesting views on this subject, highlighting a dimension that we believe cannot be overlooked in this debate, that is, the role of language, symbols, meaning and juxtapositions for the understanding of aesthetics. In this article he argues, for example, that to «isolate some *in vivo* metabolic activity pattern in response to a presumed aesthetic judgement (or whatever) will not *ipso facto* contribute useful information about the neurology of aesthetic experience» and remarks that the activation of these areas is a necessary but not sufficient condition for explaining «what is going on and why».

As he unfolds some reflections on our predisposition to representation and to communication of meaning, he draws one important thesis in the article, namely, the existence of a successive symbolic juxtaposition of the representation of our interaction with other humans and the environment, without forgetting the emotional tone, «the *prioritizing marker* attached to every cognitive object that enables an independent sorting of it with respect to other competing cognitive objects, irrespective of pattern-matching processes» and the usefulness of distinguishing «the somewhat special class of symbolically transformed emotions that are of relevance to the problem of aesthetic cognition». It means that, according to Deacon, before and below the consideration of the aesthetic experience, it is necessary to investigate the very nature of symbolic cognitive-emotional domains. That is, before and below the consideration of aesthetic experience, we treat objects as signs of something else. In this sense, the article focuses essentially on the role of juxtapositions, bisociations and blends in human cognition,

⁶ See his book *The Symbolic Species – The Co-Evolution of Language and the Brain* (1998), Norton & Company, New York, London

⁷ In *The Artful Mind - Cognitive Science and the Riddle of Human Creativity* (2006), Mark Turner (ed.), Oxford University Press, New York

claiming that we must go down to phylogenetically prior cognitive-emotional capacities as the basis of aesthetic experience.

Terrence Deacon's view is just an example, and in our opinion a relevant one, of other kinds of approaches which could be taken into account when discussing the relationship between aesthetics and the brain.

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The second section of the book begins with an article by **Chiara Cappelletto** who presents another kind of critical overview of neuroaesthetics. She identifies two substantially different approaches: one devoted to investigate the neurological condition of vision and another engaged in understanding the exchange that occurs between the spectator who perceives the world and the world that is perceived.

The first approach, epitomized by Zeki's works like *The Neurology of Kinetic Art* (1994) and *Inner Vision* (1999), is harshly criticized by Cappelletto. She points out five major problems: (1) the lack of a clear definition of the object of study; (2) the assimilation between "concept" and "perception of the object", spectator's experience and stimulus, judgment and appreciation, artwork and image; (3) the inherent fallacy of studies based on neuroimaging, accused of objectifying the subject and the first-person experience; (4) the loss of any definition of subject, through the equivalence established between artist, spectator and artwork, all reduced to mere cerebral expression; (5) the reduction of spectatorship to emotions, neglecting imagination.

So, according to Cappelletto, in Zeki's framework there would be no place for any exchange between neuroscientists and philosophers. On the contrary, the second approach, led by V.S. Ramachandran, could open the door to a dialogue for the understanding of art as embodied experience. Unlike Zeki, who relies mostly on neuroimaging data, Ramachandran takes as starting point the study of clinical cases in which the perception of the body is altered, as in the stance of phantom limb syndrome⁸. For Cappelletto, Ramachandran's work is interesting because he takes art as something dynamic and fictional that evolved as a virtual simulation of reality and therefore allows

⁸ The phantom limb syndrome occurs in people who had a limb or another part of the body amputated. The person continues to feel the missing limb and it often feels shorter and as if it was in a distorted and painful position.

experiencing situations that are unusual, unknown and innovative. She draws on different cases of Ramachandran's patients to defend that: «Art shows in a powerful way how our conscience is linked to our body and this link is just one possible degree of reversibility between effective actions and their latent possibility». In other words, art shows how our conscience is linked to our body. This observation leads Cappelletto to take in consideration also mirror neurons' theory and its contribution to neuroaesthetics. In fact, as mirror neurons are supposed to allow us to perceive what is going on in the world *as if* we were performing it, they are perfectly coherent with Ramachandran's idea of art as simulation of potentially real actions. Thus, Cappelletto concludes, the neuroaesthetic approach focused on the interaction between body and art could be particularly interesting for thinking spectatorship in performing arts theories, because it helps elaborating on the difference between “*doing as if*” and “*imagining to do as if*”.

In his article **Andrea Pinotti** distinguishes a maximalistic and a minimalistic approach to neuroaesthetics, reviewing in particular three articles: *The Neurology of Kinetic Art* (Lamb and Zeki, 1994); *The Science of Art. A Neurological Theory of Aesthetic Experience* (Ramachandran and Hirstein, 1999); and *Motion, emotion and empathy in aesthetic experience* (Freedberg and Gallese, 2007). The minimalistic approach aims at describing what goes on in our brain, at a neuronal and synaptic level, when we are engaged in an aesthetic experience. It is therefore a descriptive program. On the contrary, the maximalistic approach is prescriptive, claiming that the aim of art is to please the brain.

Even though these two programs have very different philosophical implications and rely on different epistemological premises, some neuroscientists easily shift from one to the other. It would be the case of Semir Zeki and V.S. Ramachandran. Unlike Cappelletto, who recognizes in these authors two significantly different approaches to Neuroaesthetics, Pinotti observes that although both start from a minimalistic program (the search for *constancies* in the aesthetic experience), they end up with the maximalistic claim that art *has to* titillate one or another area of the brain, since the function of art is nothing but an extension of the main function of the brain, that is, to acquire knowledge about the world.

For Pinotti, these positions – together with those overtly maximalistic of Maffei and Fiorentini, who think that the history of art mirrors the history of the brain, and of Jean-Pierre Changeux, who conceives the history of images as a real struggle for

the survival of pictorial representations – do not open a field for dialogue between neurosciences and humanities. However, even in the case of minimalistic approach, epitomized by Freedberg and Gallese, spaces for cooperation between art historians and neuroscientists are scant and the role of neurosciences in a deep understanding of the aesthetic experience is residual. Mirror neuron researches may give some valuable neurological insight on the old aesthetic issue of suspension of disbelief, however they would not scratch humanists diffidence towards trans-cultural invariants and innatism, nor the idea that aesthetic pleasure is «something exquisitely incorporeal».

Taking a radically different stand from Cappelletto and Pinotti, who consider that the relationship between neuroaesthetics and philosophy is substantially flawed by neuroaesthetics' reductionism, **Vasco Correia** claims that neurosciences may help to empirically decide a number of philosophical questions that, otherwise, «would remain mere theoretical hypotheses». Drawing on neurosciences and experimental psychology he discusses the role of emotions in aesthetic experience and defends the instrumental account of art advocated by David Hume. Neuroimaging, he warns, cannot explain the complexity of aesthetic experience, but might confirm or falsify hypotheses raised by philosophers. Thus, the encounter between philosophy and neuroaesthetics can be fruitful, since philosophers, with a long tradition of reflection upon aesthetics «provide a diverse range of hypotheses, questionings and conceptual frameworks, which may prove to be useful to neuroscientists». In turn, philosophers should be open to scientific evidence in order to clarify some philosophical questions.

In the case of the hedonic account of art, the link between beauty, pleasure and emotions is essential and Correia suggests that the results reported in Zeki's *Neural Correlates of Beauty*, together with those of other neurology and experimental biology studies, corroborate the philosophical argumentation in favour of the instrumental approach in aesthetics. Those results, he argues, may even contribute to clarify the relation between aesthetic experiences and hedonic sensation (pleasure and repulsion, both processed at the neural level by the rewarding system) and to elucidate the relation between judgements of taste and emotions (possibly entangled in the connection between orbito-frontal cortex, motor cortex and anterior cingulate).

Another somehow critical approach to neuroaesthetics is proposed by **Alexander Gerner** who defends that neither the strong conceptual nor the strong neuro

reductionist accounts of aesthetics are satisfactory, suggesting that a fruitful way to explain the aesthetic experience should encompass a conceptualization of attention. He claims that aesthetic experience can be understood as a deviation (distraction) from routine attention. In the first part of his paper he elaborates on the iconoclastic response to the art work *Who's afraid of Red, Yellow and Blue* by Barnett Newman, in order to reject the strong conceptual account of aesthetics posited by Danto, who dismisses any intrinsic difference between art objects and ordinary objects. Drawing on Seel's conceptualization of aesthetic appearing and of the interdependency of aesthetic perception and aesthetic object, Gerner suggests the existence of a fundamental difference between the aesthetic attention triggered by the aesthetic appearing and routine attention captured by non-artistic objects. Iconoclastic reactions to Newman's painting, then, are due to the failure of biological selective top-down attention mechanisms (routine attention) in categorizing the art object and processing its subject. That is, the aesthetic appearance exists before any conceptualization. That notwithstanding a merely neurological explanation of the mechanisms of attention, perception and cognition in relation to aesthetic appearing would also fail to explain the aesthetic experience. According to Gerner one of the main problems at stake is that even if the first-person experience has neural causes and neural correlates, the empirical discovery of these causes and correlates is not enough to explain the very experience. Consciousness and attention/distraction are important elements of this process and their theorization is problematic. He remarks that the concept of *routine* is an important one in describing perceptual laws in neuroaesthetics but, again, routine alone is exactly what cannot explain the aesthetic experience. Rather than claiming a theory of causal neural *correlates* of aesthetics, then, neuroaesthetics should be envisaged as a theory of neural *correspondence*. Gerner specifically criticizes Zeki's presupposition that the understanding of perceptual visual stimuli is sufficient to find a biological theory of aesthetics: he acknowledges that art objects can very well be used to study how the creative brain works, but claims that neuroaesthetics fails when it tries to explain the conscious experience of aesthetic perception. Interestingly, Gerner's conclusions cross those of Ramachandran and Hirstein: if for the neuroscientists art emerges from the exploitation and «playful and deliberate violation» (Ramachandran and Hirstein, 1999) of well-defined ordinary perceptual laws, for Gerner this *violation* is the «*necessary* starting point of research on aesthetic perception and aesthetic experience».

Finally, **Catarina Pombo Nabais** offers a different perspective on what it means thinking art in “neurological” terms. Elaborating on Deleuze’s approach to aesthetics as a philosophy of Nature, she suggests that the naturalization of art – based on Deleuze’s idea that art begins with the marks produced by territorial animals – leads to a neurological conception of art. However, “neurological” here is not reduced to a description of the brain as a specific animal organ, constituted by neurons and electrical impulses. Deleuze’s idea of “brain” is far more abstract: it is defined as «the faculty of sensation», as well as «singular spirit» and «creation of concepts, affects and percepts» and «cut of chaos». Similarly, the concept of “sensation” – which, together with that of territorial mark, plays a capital role in Deleuze’s aesthetics – is not related to bodily feelings, but to a «compound of non-human forces of the cosmos»⁹. Nevertheless, following Pombo Nabais’ reflection on the aesthetics of Deleuze and Guattari, the unifying element for Deleuze’s different “neurological” account of art emerges as an anti-humanistic program according to which the work of art exists in itself, as a «being of sensation», independently of man. Consequently, the created thing (the work of art) is autonomous from the spectator and from the artist who created it.

If the aim of art is the pure being of sensation, Pombo Nabais notices, then aesthetics is transformed into «an ontology of pure sensations». In this case, «the goal of art can then be defined as anti-humanistic, since it corresponds to the extraction of all the subjective features of the sensation». And, while the anti-humanism of neurosciences erases the subject in favour of neural activity and evolutionary pressure, both related to the body, Pombo Nabais argues that Deleuze and Guattari push even more their anti-humanistic stance claiming it to a vitalistic cosmological program in which art «captures the intensity of life’s immanence», since it would result from «a transcendental empiricism of the sensation as an inorganic exercise of the brain».

Marco Pina, Olga Pombo, Silvia di Marco

⁹ Deleuze, G. and Guattari, F. (1994) *What is Philosophy?*, Verso, London-New York. Cited in Pombo Nabais’ article.